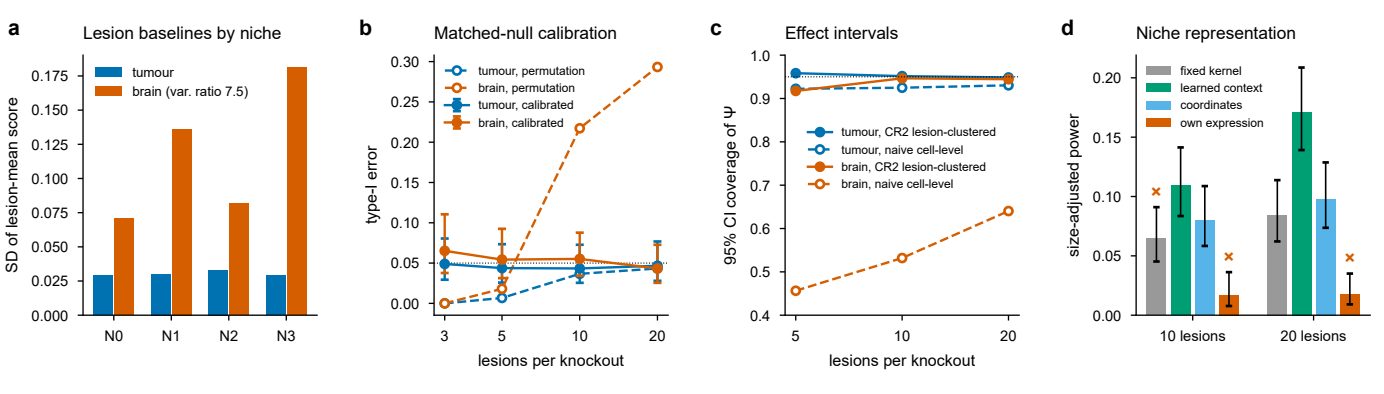
a
Lesion baselines by niche
tumour
brain (var. ratio 7.5)
SD of lesion-mean score
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
N0
N1
N2
N3
b
Matched-null calibration
tumour, permutation
brain, permutation
tumour, calibrated
brain, calibrated
type-I error
0.30
0.25
0.20
0.15
0.10
0.05
0.00
3
5
10
20
lesions per knockout
c
Effect intervals
tumour, CR2 lesion-clustered
tumour, naive cell-level
brain, CR2 lesion-clustered
brain, naive cell-level
95% CI coverage of Ψ
1.0
0.9
0.8
0.7
0.6
0.5
0.4
5
10
20
lesions per knockout
d
Niche representation
fixed kernel
learned context
coordinates
own expression
size-adjusted power
0.20
0.15
0.10
0.05
0.00
10 lesions
20 lesions